

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

A Subsidiary of Oil and Natural Gas Corporation (ONGC)

Kuthethoor P.O., Via Katipalla, Mangalore - 575030, Karnataka, India

STANDARD OPERATING PROCEDURE

OPERATION, INSPECTION & OVERPRESSURE INTERVENTION FOR CENTRIFUGAL
HEAVY GASOIL BOOSTER PUMPS

Conforming to API 610 (12th Edition) & OISD-STD-119 Guidelines

Document Number:	MRPL - SOP - PUMP - 610 - R4	Effective Date:	2026-01-15
Revision Number:	Rev 4.2 (Annual Review)	Review Cycle:	Annual (Mandatory)
Process Unit:	Unit 01 — Heavy Gasoil Hydrotreater (HCU-01)	Plant Area:	Pumping Station Manifold Block 3
Equipment Tags:	P-101A (Primary) / P-101B (Standby)	Criticality Rating:	CATEGORY A — MISSION CRITICAL
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1.0 EQUIPMENT SCOPE & DESIGN SPECIFICATIONS

Centrifugal pumps **P-101A** and **P-101B** provide primary pressurized feedstock to Catalytic Hydrotreater **Reactor-B** via discharge line **12-CDU-401-HC**. The pumps handle hot, hydrogen-saturated heavy atmospheric gasoil containing up to 2.2 wt% organic sulfur at elevated hydraulic head.

Parameter	Engineering Value	Design Standard / Compliance
Pump Classification	Centrifugal Between-Bearing, Radially Split (Type BB2)	API 610 12th Edition
Manufacturer / Model	Sulzer Pumps India / Model MSD-150-380	OEM Technical Manual SM-904
Fluid Service	Heavy Atmospheric Gasoil (HAGO / HC Feed)	Specific Gravity: 0.884 at 60°C
Operating Pumping Capacity	380.0 m³/h (1,673 US GPM) Nominal	Best Efficiency Point (BEP) 84.5%
Differential Head	142.0 meters (465.8 feet)	Shut-off Head: 178.0 meters
Suction Design Pressure	15.0 to 25.0 PSI (1.03 to 1.72 bar gauge)	NPSHa: 6.8 m (NPSHr: 4.2 m)
Normal Operating Discharge Pressure	80.0 to 120.0 PSI (5.51 to 8.27 bar gauge)	Normal continuous operation envelope
Maximum Allowable Working Pressure (MAWP)	500.0 PSI (34.47 bar gauge)	Hydrostatic Shop Test: 750 PSI (1.5x)
Automatic Overpressure Trip Limit	450.0 PSI (31.02 bar gauge)	Instrument Tag: PT-101 / Interlock I-101
Prime Mover / Electric Motor	450 kW (600 HP) Induction, 3-Phase 6.6 kV	Zone 1, Ex-d IIC T4 Flameproof
Operating Speed	2,980 RPM at 50 Hz	VFD Frequency: 45.0 - 52.0 Hz
Mechanical Seal Arrangement	Dual Pressurized Cartridge Seal with Plan 53B	API 682 Category 2 / Type A
Seal Barrier Fluid System	Synthetic Polyalphaolefin (PAO) at 2.0 bar delta	Cooling loop with finned air exchanger

2.0 SAFE OPERATIONAL BOUNDARIES & INSTRUMENT TRIP MATRIX

Operators and control room panel technicians must maintain process parameters strictly within the green continuous boundary. Operating outside safe thresholds initiates graduated supervisory escalation and emergency shutdowns.

Parameter / Sensor Tag	Normal Range	Low Alarm (LL)	High Alarm (HI)	Emergency Trip (HH)	Safe Corrective Action
Suction Pressure (PT-102)	15.0 - 25.0 PSI	12.0 PSI	30.0 PSI	8.0 PSI (LL)	Check feed surge tank level
Discharge Pressure (PT-101)	80.0 - 120.0 PSI	65.0 PSI	140.0 PSI	450.0 PSI	Emergency relief valve SV-402 opens
Inboard Bearing Temp (TT-204)	50.0 - 70.0 °C	--	85.0 °C	95.0 °C	Check lube oil pressure and flow
Outboard Bearing Temp (TT-205)	50.0 - 72.0 °C	--	85.0 °C	95.0 °C	Lube cooler fan verification
Shaft Vibration Horiz (VT-102A)	1.8 - 3.2 mm/s	--	4.5 mm/s	7.0 mm/s	Perform FFT spectrum analysis
Shaft Vibration Vert (VT-102B)	1.5 - 2.8 mm/s	--	4.5 mm/s	7.0 mm/s	Inspect coupling alignment
Motor Stator Amperage (IT-101)	165.0 - 195.0 A	90.0 A	215.0 A	245.0 A	Verify valve stiction downstream
Seal Pot Differential (PDT-301)	2.0 - 2.5 bar	1.5 bar	3.2 bar	1.0 bar (LL)	Isolate and vent seal reservoir

AUTOMATIC EMERGENCY TRIP LOGIC (INTERLOCK I-101):

- Logic Circuit: 2-out-of-3 (2oo3) voting on discharge pressure transmitter **PT-101**.
- Primary Action: Instant trip signal dispatched to 6.6 kV Vacuum Circuit Breaker (VCB-101A) within 45 ms.
- Secondary Safety Action: Pneumatic actuator on Emergency Relief Valve **SV-402** energizes open, routing pressurized vapor inventory into the high-pressure **Flare-Header**.
- Interlock Reset Requirement: Manual reset at field local control panel (LCP) forbidden until downstream pressure drops below 200 PSI and Dual Supervisor Authorization is registered.

3.0 PRE-START INSPECTION & LINE-UP CHECKLIST

Before energizing the drive motor on Pump P-101A following a maintenance outage or shift turnover, the field operator must complete and physically sign the following 10-point mechanical inspection checklist:

#	Inspection Point	Acceptable Condition	Initial / Verified
1	Bearing Housing Lube Oil Level	Between 1/2 and 2/3 of sight bullseye glass. Color: Amber, clear, no water haze.	☐ Operator Sam
2	Cooling Water Circulation	Inlet pressure > 3.0 bar. Flow sight glass rotor spinning freely. Outlet temp < 35°C.	☐ Operator Sam
3	Suction Manual Block Valve (MOV-101)	100% Full Open and car-sealed open (CSO). Stem thread inspected.	☐ Operator Sam
4	Discharge Manual Block Valve	Throttle valve cracked 10% open for priming. Full discharge block valve closed.	☐ Operator Sam
5	Casing Vent & Priming Bleed	Vent valve opened to closed drain until steady hydrocarbon liquid stream observed.	☐ Operator Sam
6	Shaft Free-Spin Hand Check	Shaft rotated 2 full revolutions by hand through coupling guard inspection port.	☐ Operator Sam
7	Plan 53B Seal Barrier Pressure	Accumulator nitrogen precharge 18 bar. PAO fluid pressure 22 bar (> suction + 2 bar).	☐ Operator Sam
8	Minimum Flow Spillback Valve (FV-101)	Verified operational on auto-bypass. Recirculation line unblocked.	☐ Operator Sam
9	Instrument Purge & Tag Status	PT-101, TT-204, VT-102 transmitters active on SCADA with zero active diagnostic alarms.	☐ Control Room
10	Work Permit & Lockout-Tagout (LOTO)	Cold/Hot work permits closed. Electrical isolator tags de-isolated by certified electrician.	☐ Supervisor Jane

4.0 NORMAL STARTUP, COMMISSIONING & FLOW RAMP-UP SEQUENCE

Step 1 — Field Readiness Authorization: Confirm field operator is standing by at Local Control Panel (LCP-101) with two-way radio link to the main control room DCS console.

Step 2 — Energize Seal Support System: Verify barrier fluid loop PDT-301 differential is stable at 2.2 bar gauge. Ensure auxiliary lube oil pump (P-101-AUX) is running on automatic recirc.

Step 3 — Open Suction Manifold: Slowly open the 14-inch suction isolation gate valve. Confirm suction pressure on PT-102 rises to 18–22 PSI without hydraulic hammer.

Step 4 — Motor Breaker Close: DCS panel operator depresses 'START P-101A'. High-voltage VCB contactor closes. Motor inrush current peaks at ~950 A for 3.2 seconds before settling to 175 A idle load.

Step 5 — Monitor Immediate Acoustic & Vibration Response: Field operator inspects casing within 10 seconds. If cavitation rumble, rubbing, or violent shaking (> 5.0 mm/s) occurs, trip immediately.

Step 6 — Verify Pressure Rise: Discharge pressure transmitter PT-101 must register immediate pressure rise to shutoff head (~165 PSI) against the closed discharge valve.

Step 7 — Throttled Valve Ramp: Over a duration of 45 to 60 seconds, gradually open the 12-inch discharge motor-operated valve (MOV-102) to establish process flow into Reactor-B.

Step 8 — Establish Operating Point: Adjust discharge control valve to achieve rated 380.0 m³/h flow rate. Verify discharge pressure settles between 95.0 and 115.0 PSI.

Step 9 — Thermal Equilibrium Monitoring: Log bearing temperatures TT-204 every 5 minutes for the first 30 minutes. Temperature must plateau below 72.0 °C.

Step 10 — Transfer Standby Pump to Auto-Start: Switch identical twin booster pump P-101B to 'AUTO STANDBY' mode. If P-101A trips, P-101B will auto-start within 4.5 seconds.

5.0 EMERGENCY INTERVENTION PROTOCOL & FOUR-EYES SUPERVISORY GATE

5.1 Immediate Automated Response to Overpressure Surge:

If discharge pressure exceeds **450.0 PSI (31.02 bar)** or bearing temperature breaches **95.0 °C**:

1. The automated safety interlock trips breaker VCB-101A, de-energizing the main 450 kW drive motor.
2. Emergency Safety Relief Valve **SV-402** opens fully, venting trapped hydrocarbon volume into the **Flare-Header**.
3. Minimum recirculation bypass valve opens 100% to dissipate residual hydraulic momentum.

5.2 Strict Four-Eyes Principle for Operational Interventions:

In accordance with MRPL Safety Rulebook Part IV and IEC 62443 industrial cybersecurity standards, **no single human operator or autonomous AI agent may re-energize equipment or reset an emergency trip unilaterally.**

MANDATORY DUAL-AUTHORIZATION WORKFLOW:

Step	Authorized Role	Required Verification Criteria	System Enforcement Gate
Stage 1: Verification	Shift In-Charge / Supervisor (Jane)	Review live telemetry on PT-101 (< 200 PSI), inspect field leak report, verify motor winding resistance.	Requires valid Supervisor JWT token. Transitions approval request to STAGE_1_VERIFIED. Engine remains safely PAUSED.
Stage 2: Final Authorization	Plant Authorizer / VP Operations (Rohit)	Confirm downstream reactor catalyst bed thermal gradient is safe. Approve operational component restart tool call.	Requires distinct Authorizer JWT token (cannot match Stage 1 user). Executes restart_component tool and logs SHA-256 audit signature.

WARNING: Attempting to bypass the Four-Eyes authorization gate constitutes a Level-1 Process Safety Incident (PSI) under OISD-156 and triggers immediate disciplinary investigation.

6.0 PREVENTIVE MAINTENANCE & RELIABILITY INSPECTION SCHEDULE

Frequency	Component / Subsystem	Inspection Task & Acceptance Tolerance	Lubricant / Consumable
Daily (Every Shift)	Pump Bearing Housings	Visual check of oil level in constant level oiler. Log bearing skin temp (< 75°C).	ISO VG 46 Synthetic Turbine Oil
Weekly	Shaft Seal Pot 53B	Record nitrogen bladder pressure. Verify PAO barrier fluid level. Refill if < 40%.	Mobil SHC 524 PAO Synthetic
Monthly	Flexible Spacer Coupling	Laser alignment check: Angular misalignment < 0.05 mm, Parallel offset < 0.05 mm.	High-speed synthetic grease NLGI 2
Quarterly (90 Days)	Lube Oil Sampling & Assay	Draw 250 ml oil sample for lab spectroscopy. Viscosity within ±10%, TAN < 0.5 mg KOH/g.	Laboratory Testing (ASTM D445)
Biannual (180 Days)	Suction Strainer Basket	Differential pressure across basket < 0.35 bar. Clean debris and flush mesh with kerosene.	Stainless Steel 40-mesh element
Annual Turnaround	Full Casing & Impeller Overhaul	Measure wear ring diametral clearance (0.35 - 0.45 mm). Dye penetrant on impeller vanes.	API 610 12% Chrome Stainless Spares

7.0 CHEMICAL HAZARDS (H2S / TOXIC GASES) & ESCALATION MATRIX

7.1 Hydrogen Sulfide (H2S) & Toxic Sour Gas Exposure:

Heavy atmospheric gasoil contains dissolved H2S in concentrations up to 1,800 ppmw in the feed drum. Any mechanical seal leak or piping breach generates lethal gaseous H2S clouds.

- OSHA PEL: 20 ppm ceiling | NIOSH IDLH: 100 ppm.
- Personal Multigas Detector (PGM-1100) mandatory before approaching P-101A.
- If ambient H2S alarm sounds (> 10 ppm), immediately don positive-pressure SCBA and evacuate upwind to Assembly Point Charlie.

7.2 Incident Escalation & Control Room Directory:

Emergency Role	Designation	Contact Phone / Speed Dial	Radio Channel
Main Control Room (MCR-01)	Shift In-Charge Operations	Ext: 4401 / 4402	Channel 01 (Ops)
Fire & Process Safety Station	Refinery Fire Marshal	Ext: 101 / 9911 (Emergency Hot Line)	Channel 09 (Fire)
Rotating Equipment Specialist	Reliability Engineering Dept	Ext: 4820 / 4825	Channel 04 (Mech)
Occupational Health Center (OHC)	Chief Medical Officer	Ext: 3300 / 3301	Channel 07 (Med)

DOCUMENT VALIDATION & ENGINEERING ENDORSEMENT:

Certified that this Standard Operating Procedure conforms to Mangalore Refinery and Petrochemicals Limited Safety Management System (SMS) Manual Volume III and OISD Standard 119.

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